

٢٥٠

3rd year

١ شارات ونظم محلي ٢

Signals & Systems Lab

ذکاء ٥٦٥

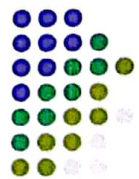
Lab (7)

Transformations of Signals [Discrete Signals]

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Transformations of the Time Variable for Discrete-Time Signals



$x[n]$ discrete-time signal n_0 integer number

- **Time shifting.** $y[n] = x[n - n_0] \Rightarrow$ $n_0 > 0$ shifted by n_0 units (or samples) to the right
 $n_0 < 0$ shifted by n_0 units to the left
`stem(n+n0, x)`

- **Time reversal** $y[n] = x[-n] \Rightarrow$ $y[n]$ is the reflected (about the vertical axis) signal.
`stem(-n, x)`

downsampling
 a must be a positive integer

- **Time scaling.** $y[n] = x[an] \Rightarrow$
upsampling
 $1/a$ must be a positive integer

$a > 1$ and $a \in \mathbb{Z} \Rightarrow$ **downsampling or decimation**
 results in time compression of the signal

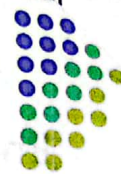
`y = downsample(x, a)`
`y = x(1:a:end)`

$0 < a < 1 \Rightarrow$ time-expanded version of $x[n]$
 $(1/a) - 1$ zeros are inserted between two consecutive samples of $x[n]$

upsampling time-expansion operation
`y = upsample(x, 1/a)`
`y(1:1/a:end) = x`

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Transformations of the Time Variable for Discrete-Time Signals



The downsampling and upsampling processes are illustrated in the next example for the discrete-time signal $x[n] = [1, 2, 3, 4, 5, 6]$, $0 \leq n \leq 5$.

```
x = [1, 2, 3, 4, 5, 6]
```

```
x = 1 2 3 4 5 6
```

```
a = 2;
```

```
xds = downsample(x, a)
```

```
xds = 1 3 5
```

```
xds = x(1:a:end)
```

```
xds = 1 3 5
```

```
a = 1/2;
```

```
xups = upsample(x, 1/a)
```

```
xups = 1 0 2 0 3
      0 4 0 5 0 6 0
```

```
xups = zeros(1, 1/a * length(x))
```

```
xups(1:1/a:end) = x
```

```
xups = 1 0 2 0 3
      0 4 0 5 0 6 0
```

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Transformations of the Time Variable for Discrete-Time Signals



Example

Consider the sequence $x[n] = 0.9^n$, $-10 \leq n \leq 10$. Plot the sequences $x[n-10]$, $x[n+10]$, $x[3n]$, $x[n/3]$, and $x[-n]$.

```
n = -20:20;
```

```
p = (n >= -10) & (n <= 10);
```

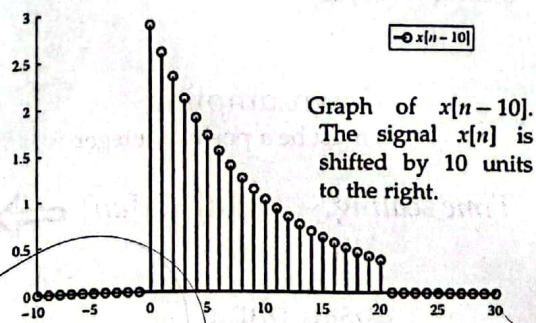
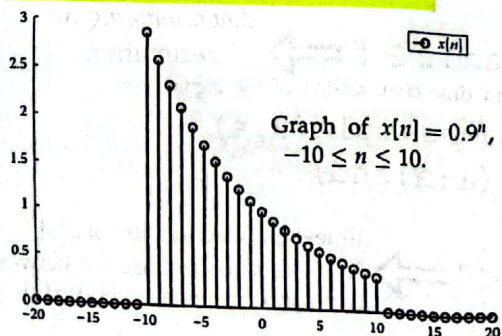
```
x = (0.9.^n) .* p;
```

```
stem(n, x);
```

```
legend('x[n]');
```

```
stem(n+10, x)
```

```
legend('x[n-10]')
```



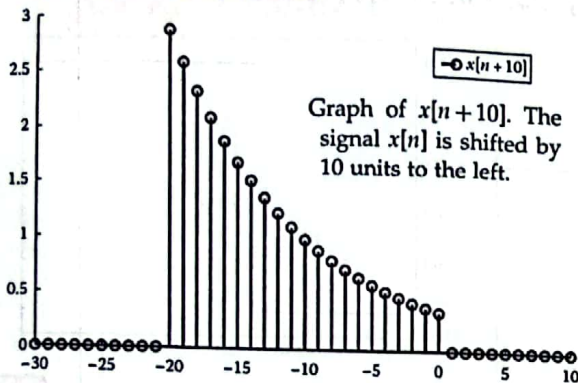
4

Transformations of the Time Variable for Discrete-Time Signals

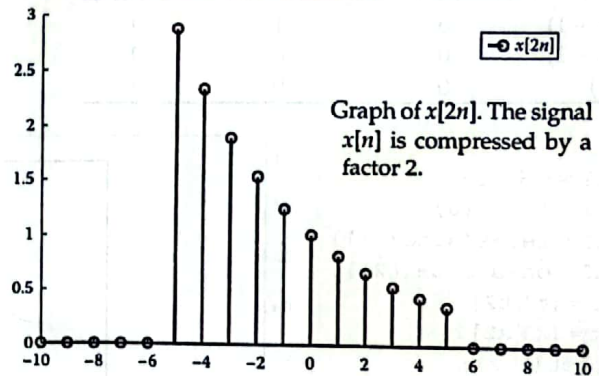
Example

Consider the sequence $x[n] = 0.9^n$, $-10 \leq n \leq 10$. Plot the sequences $x[n-10]$, $x[n+10]$, $x[3n]$, $x[n/3]$, and $x[-n]$.

```
stem(n-10, x)
legend('x[n+10]')
```



```
a = 2;
xd = downsample(x, a)
nd = -10:10;
stem(nd, xd);
legend('x[2n]')
```



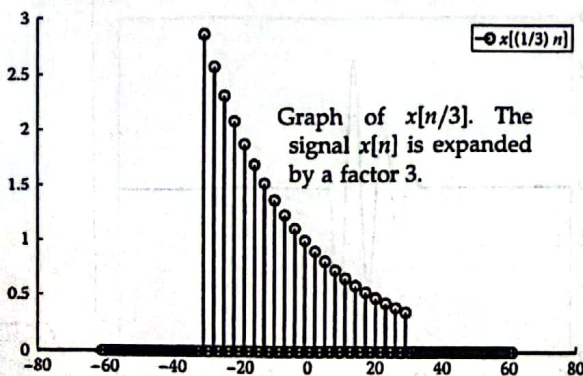
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Transformations of the Time Variable for Discrete-Time Signals

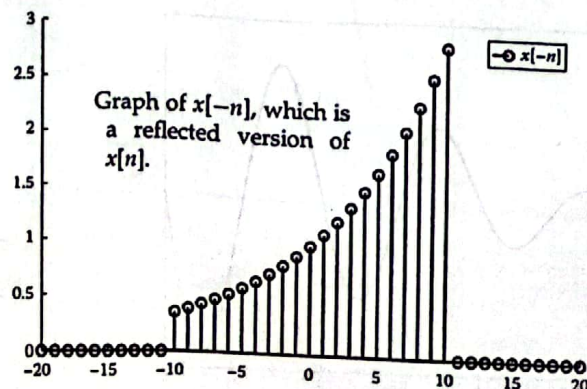
Example

Consider the sequence $x[n] = 0.9^n$, $-10 \leq n \leq 10$. Plot the sequences $x[n-10]$, $x[n+10]$, $x[3n]$, $x[n/3]$, and $x[-n]$.

```
a = 1/3
xup = upsample(x, 1/a)
nup = -61:61
stem(nup, xup)
legend('x[(1/3)n]')
```



```
stem(-n, x)
legend('x[-n]')
```



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Solved Problems

Problem 2 Plot the signal $x(t) = u(t+1)u(t-1)$

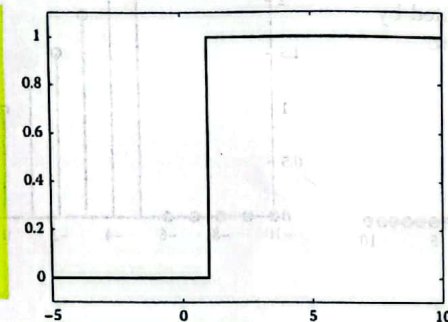
- Without using the command heaviside
- Using the command heaviside

Solution

	$-\infty < t < -1$	$-1 \leq t < 1$	$1 \leq t < \infty$
$u(t+1)$	0	1	1
$u(t-1)$	0	0	1
$x(t)$	0	0	1

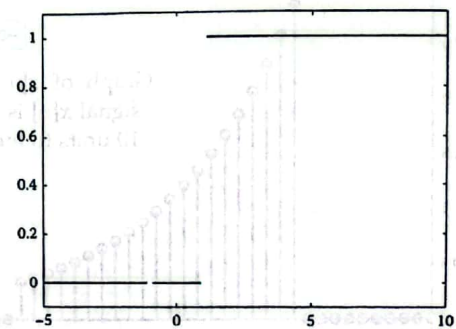
a.

```
t1=-5:.1:1;
t2=1:.1:10;
x1=zeros(size(t1));
x2=ones(size(t2));
t=[t1 t2];
x=[x1 x2];
plot(t,x);
ylim([-0.1 1.1]);
```



b.

```
t=-5:.1:10;
x=heaviside(t+1).*heaviside(t-1);
plot(t,x);
ylim([-0.1 1.1]);
```



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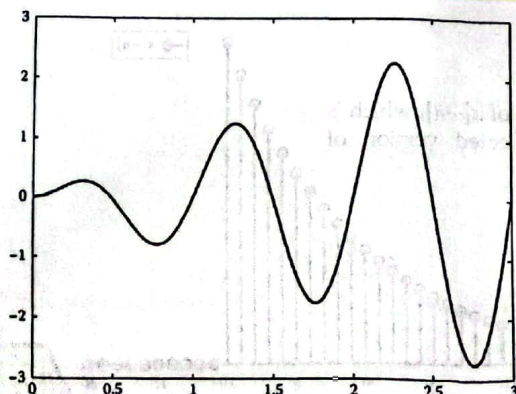
Solved Problems

Problem 4 Plot the signal $x(t) = t \sin(2\pi t)(u(t) - u(t-3))$.

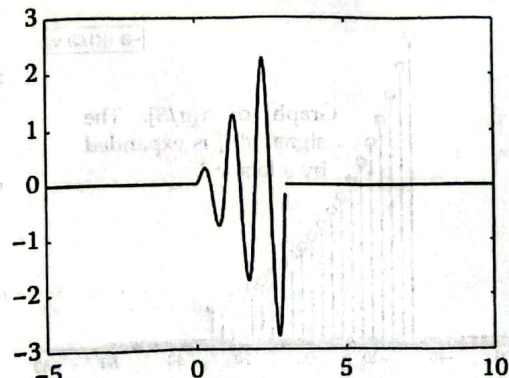
Solution

$$x(t) = \begin{cases} t \sin(2\pi t), & 0 \leq t \leq 3 \\ 0, & t < 0 \text{ and } t > 3 \end{cases}$$

```
t=0:.01:3;
x=t.*sin(2*pi*t);
plot(t,x)
```



```
t=-5:.01:10;
x=t.*sin(2*pi*t).*(heaviside(t)-heaviside(t-3));
plot(t,x)
```



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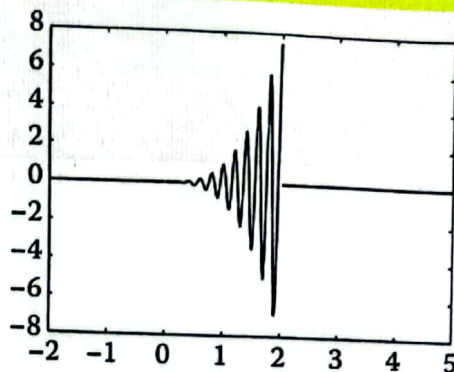
Solved Problems

Problem 5 Plot the signal $x(t) = t^3 \cos(10\pi t) p_2(t-1)$, where $p_T(t)$ is a rectangular pulse of duration T .

Solution

$$p_2(t-1) = u(t-1+2/2) - u(t-1-2/2) = u(t) - u(t-2).$$

```
t=-2:.01:5;
x=(t.^3).*cos(10*pi*t).*(heaviside(t)-heaviside(t-2));
plot(t,x)
```



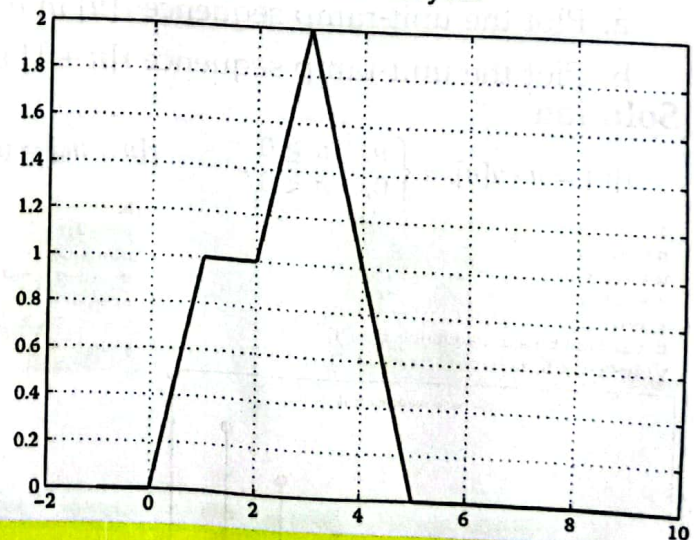
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Solved Problems

Problem 7 Express the signal depicted in the figure as a sum of ramp functions. Plot your result for verification.

Solution

$$x(t) = r(t) - r(t-1) + r(t-2) - 2r(t-3) + r(t-5).$$



```
t=-1:.001:10;
x=t.*heaviside(t)-(t-1).*heaviside(t-1)+(t-2).*heaviside(t-2)-
2*(t-3).*heaviside(t-3)+(t-5).*heaviside(t-5);
plot(t,x);
grid;
```

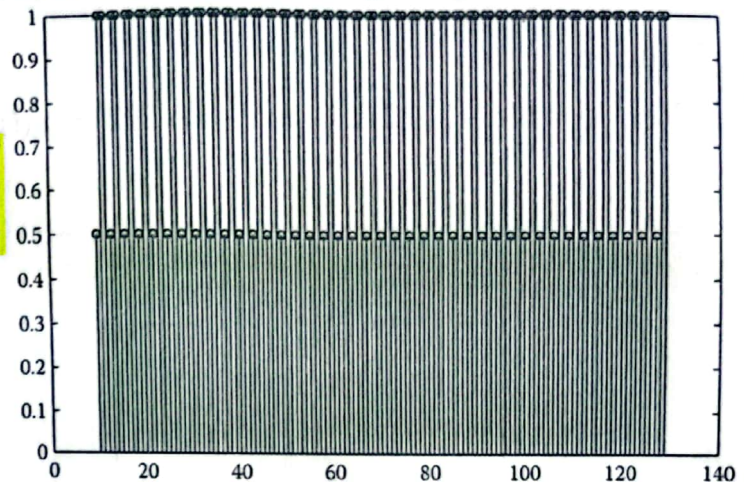
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Solved Problems

Problem 8 Draw the periodic discrete-time signal $x[n]$ that is depicted in the figure. The signal $x[n]$ in one period is given by $x[n] = [0.5, 1, 1]$. Notice that the signal begins at the time instance $n = 10$.

Solution

```
x = [0.5 1 1];
xp = repmat(x, 1, 40);
stem(10:length(xp)+9, xp);
```



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Solved Problems

Problem 11

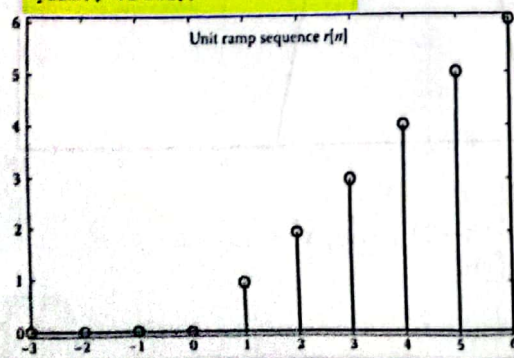
- Plot the unit-ramp sequence $r[n]$ in the time interval $-3 \leq n \leq 6$.
- Plot the unit-ramp sequence $r[n+1]$ in the same time interval.

Solution

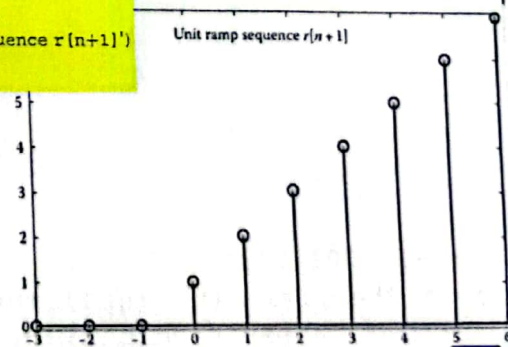
$$r[n] = n \cdot u[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

$$r[n - n_0] = (n - n_0)u[n - n_0] = \begin{cases} n - n_0, & n \geq n_0 \\ 0, & n < n_0 \end{cases}$$

```
a.
n = -3:6;
u = (n >= 0);
r = n.*u;
stem(n, r)
title('Unit ramp sequence r[n]')
ylim([-1 6.1])
```



```
b.
n = -3:6;
u1 = (n >= -1);
r = (n+1).*u1;
stem(n, r)
title('Unit ramp sequence r[n+1]')
ylim([-1 7.1])
```



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